

Derrin Bright

Bioinformatics | Biotechnology | ML in Biology

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Education

National University of Singapore (NUS) <i>Master of Science in Precision Health and Medicine, Yong Loo Lin School of Medicine</i>	Singapore Aug 2026-Present
Vellore Institute of Technology (VIT) <i>Bachelor of Technology in Biotechnology</i>	CGPA: 8.8 Vellore, India 2022-2026

Technical Skills

Computational Biology Tools	BLAST AlphaFold AutoDock Vina GROMACS MGLTools Bioconductor OpenBabel Cytoscape Seurat GSEA FIMO
Data Analysis Tools	Neo4J KNIME NetworkAnalyst
Programming Tools	R Python
Machine Learning Tools	TensorFlow Keras NumPy Pandas Matplotlib
Certificates	Machine Learning A-Z Python for Data Science, AI & Development Bulk RNA-Seq Data Analysis Advanced Bioinformatics Single-Cell RNA-Seq Data Analysis using R & Python

Research Experiences

Project Intern at Nanyang Technological University, Singapore - Developing a transcription factor activity inference pipeline on large-scale postmortem brain RNA-seq data to investigate gene regulatory dynamics across neurodegenerative disease progression. - Supervised by Dr. Guillaume Thibault Guillaume Thibault Lab	Jan 2026 - June 2026
Research Intern at Boston University, USA - Developed an automated bioinformatics pipeline for transcription factor motif analysis across 200+ HPV genomes, identifying differentially enriched TFs between high-risk and low-risk types. - Integrated motif-based approaches with deep learning sequence models to assess regulatory activity and characterise binding site conservation across 1,000+ clinical isolates. - Supervised by Dr. Juan Fuxman Bass	Sept 2025 - April 2026
Research Intern at University of Liverpool, UK - Analyzed single-cell RNA-seq data from CLL and Richter's Transformation patients, developing machine learning models (XGBoost, SVM, Random Forest) to differentiate disease states at the transcriptomic level. - Built genome-scale metabolic models (GEMs) for CLL and Richter's Transformation cell types using the ftnIT algorithm, and identified differential metabolic reactions and enriched pathways between disease states and cell types. - Supervised by Dr. Francesco Del Carratore	Sept 2025 - August 2026
Research Intern at IIT Bombay, India - Conducted differential expression analysis and Weighted Gene Co-expression Network Analysis (WGCNA) to identify key gene modules associated with autoimmune disease progression, followed by Gene Set Enrichment Analysis (GSEA) and Protein-Protein Interaction (PPI) network construction using NetworkAnalyst to uncover dysregulated pathways and functional gene relationships. - Built and evaluated predictive models, including Logistic Regression, SVM, XGBoost, and Deep Neural Networks, to classify disease vs. healthy states based on gene expression, enhancing biological interpretability through integrated network and machine learning approaches. - Supervised by Dr. Prakriti Tayalia	May 2025 - Oct 2025
Research Intern at Ashoka University, India - Analyzed five assembled E. coli genomes for carbapenem and beta-lactam resistance using BLAST and RGI-CARD to identify mutations, and correlated genotype-phenotype relationships through AST report analysis. - Performed Multiple Sequence Alignment (MSA) of Penicillin-Binding Proteins (PBPs) from the five E. coli genomes against a reference wild-type sequence to detect conserved mutations, and conducted genomic analysis that led to the identification of a 12-base pair insertion in the PBP3 gene, contributing to antibiotic resistance mechanisms. - Automated genomic data processing by developing a custom script to systematically rename and organize FASTA files, creating a structured dataset for phylogenetic analysis using the PhaME tool. - Supervised by Dr. Shraddha Karve	Jan 2025 - Mar 2025
Biotech Team Lead at NovaLinks, India - Led a project integrating miRNA analysis and histopathological imaging with machine learning to enhance lung cancer subtype classification and improve diagnostic precision. - Conducted research on miRNA biomarkers specific to lung cancer and provided biological insights to support the development of an accurate classification model for distinguishing adenocarcinoma from squamous cell carcinoma.	Feb 2025 - April 2025

Publications

Comparative Analysis of Transcription Factor Binding Sites in the Long Control Region Across Human Papillomavirus Types | *Viruses*, 2026, 18(6):646 (MDPI)

Integrative transcriptomic analysis identifies predictive gene signatures across autoimmune diseases: implications for personalized therapeutic targeting | *Under review at Journal of Autoimmunity (Elsevier)*

Comparative Analysis of HPV-Positive Cervical and Head and Neck Cancers Using WGCNA and Machine Learning Identifies Shared Biomarkers | *Under review at Tumour Virus Research (Elsevier)*

HPV-Associated Transcriptional Landscape in Cervical and Head and Neck Cancers | *In revision at Infectious Agents and Cancer (Springer Nature)*

Technical Projects

Transcriptomic Comparison of HPV-Associated Cancers Using Co-Expression Networks

- Conducted a comparative analysis of HPV-positive cervical and head and neck cancers using TCGA data. Performed DESeq2 and WGCNA to identify shared genes between both cancers.
- Applied machine learning models to filter features, identifying 16 HPV signature genes, and validated findings using single-cell RNA-seq analysis.
- Constructed PPI networks and performed functional enrichment analyses to identify shared molecular mechanisms underlying HPV-driven tumorigenesis.

Computational Identification of Antimalarial Leads through Fingerprint Similarity Analysis

[GitHub Link](#)

- Screened a library of antimalarial molecules against a query (chloroquine drug) by generating molecular fingerprints and computing Tanimoto similarity coefficients using KNIME and RDKit nodes.
- Extracted molecular data from the Protein Data Bank (PDB) and visualized pairwise similarity scores as a heatmap to identify high-similarity candidates for antimalarial drug development.

In Silico Analysis of 3DTC/CEP-6331 Complex using Molecular Docking

[GitHub Link](#)

- Implemented a molecular docking simulation using AutoDock Vina to model the interaction between the 3DTC kinase and the CEP-6331 inhibitor, achieving a predicted binding affinity of -12.23 kcal/mol.
- Analyzed the top binding pose to identify the specific amino acids responsible for the stable interaction, including key hydrophobic (ILE180, LEU274) and polar (LYS228, ASP284) residues.

Single-Cell Transcriptomic Profiling of the B16 Melanoma's Microenvironment

[GitHub Link](#)

- Processed raw single-cell transcriptomic data from a B16 melanoma model using R (Seurat), applying quality control, normalization, and clustering to map the tumor's cellular heterogeneity.
- Characterized the resulting cell clusters by annotating fourteen distinct cell types within the tumor microenvironment and performed differential expression analysis to identify their unique gene signatures.

Honors and Leadership

First runner-up in the Biomimicry Innovation Challenge and Showcase, Vellore Institute of Technology, 2024

Top 10 Winner of Bio-Inspired Design Fest (BIDFEST) Ideathon, Vellore Institute of Technology, 2024

Vice Chairperson at Alpha Bio Cell, VIT University, India

January 2024 - December 2024

- Led several biotechnology and bioinformatics-related events, including hackathons, workshops, and quizzes, enhancing the club's engagement.
- Mentored junior core members and applied leadership, technical, and project management skills to ensure successful event planning and execution.

Extra-Curricular

Volunteer | National Service Scheme, NGO

Conducted sessions in government schools on career opportunities and the importance of joining the armed forces.

Volunteer | Becoming I Foundation, NGO

Taught science, math, and english in government schools, supporting underprivileged communities.

Event Coordinator | BioSummit, School of Biosciences and Technology

Organized a flagship event connecting academia and food industries, managing food distribution and ensuring smooth execution.